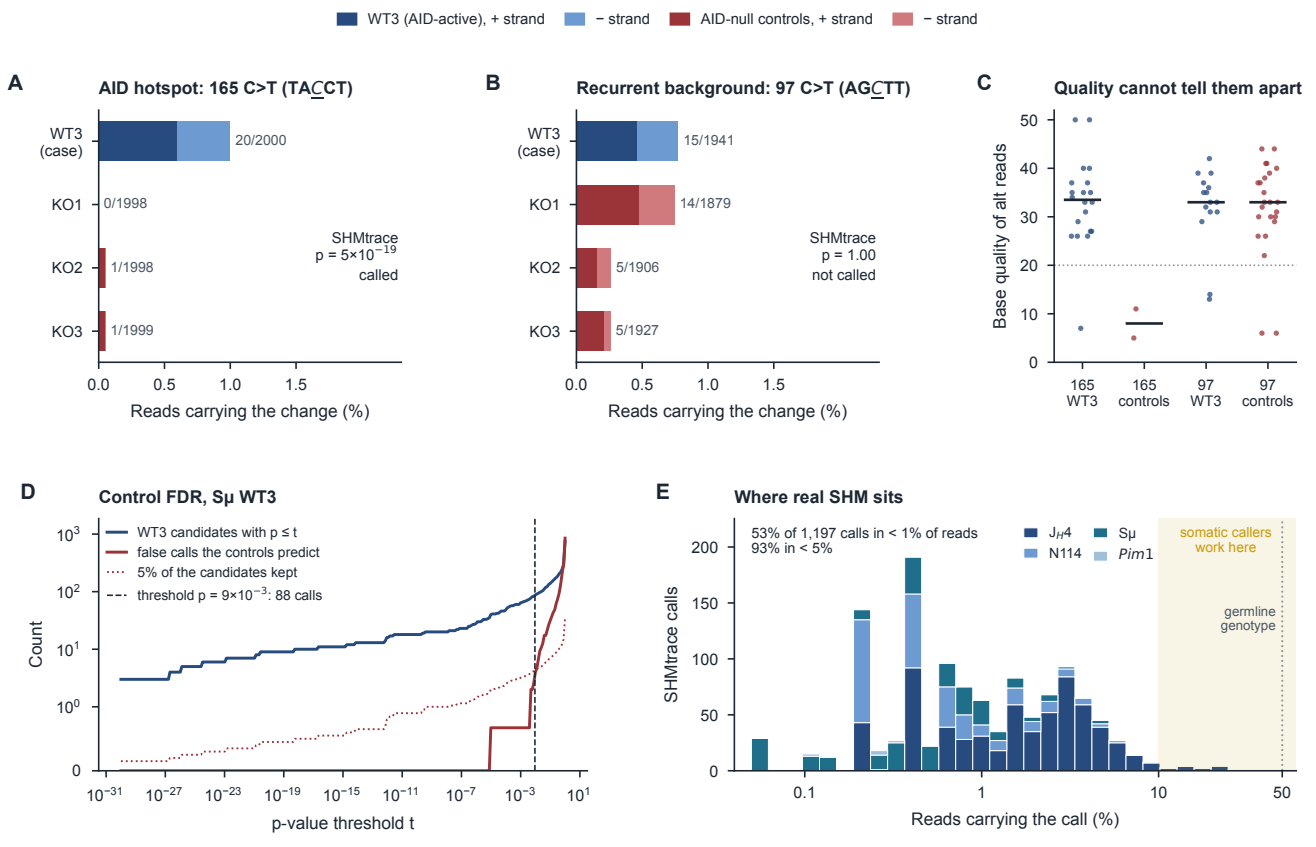

WT3 (AID-active), + strand
− strand
AID-null controls, + strand
− strand
A
AID hotspot: 165 C>T (TACCT)
WT3 (case)
KO1
KO2
KO3
20/2000
0/1998
1/1998
1/1999
SHMtrace
p = 5×10−19
called
0.0
0.5
1.0
1.5
Reads carrying the change (%)
B
Recurrent background: 97 C>T (AGCTT)
WT3 (case)
KO1
KO2
KO3
15/1941
14/1879
5/1906
5/1927
SHMtrace
p = 1.00
not called
0.0
0.5
1.0
1.5
Reads carrying the change (%)
C
Quality cannot tell them apart
Base quality of alt reads
0
10
20
30
40
50
165 WT3
165 controls
97 WT3
97 controls
D
Control FDR, Sμ WT3
WT3 candidates with p ≤ t
false calls the controls predict
5% of the candidates kept
threshold p = 9×10−3: 88 calls
Count
10^0
10^1
10^2
10^3
10−31
10−27
10−23
10−19
10−15
10−11
10−7
10−3
101
p-value threshold t
E
Where real SHM sits
53% of 1,197 calls in < 1% of reads
93% in < 5%
JH4
Sμ
N114
Pim1
somatic callers work here
germline genotype
SHMtrace calls
0
50
100
150
200
0.1
1
10
50
Reads carrying the call (%)